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Waves on a String

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Abstract

Stationary waves are formed when two waves of equal speed and frequency superpose while travelling in different directions. In this experiment, stationary waves were created on a string by using an electronic vibration generator connected to a pulley system and an oscilloscope, from which the frequency was determined when each harmonic had formed. The formula $v = f\lambda$ was then used to find the speed of the wave, from which another equation $v = \sqrt{(T/\mu)}$ was used to identify the mass of the string per unit length. The wave speed found was $v = 18.6 \pm 0.5$ ms^{-1} and $\mu = 8.51 \pm 0.22$ gm^{-1}, while the true value of μ was 8.68, which lied within our error bounds and differs from our result by only 2%.

Symbols and Equations

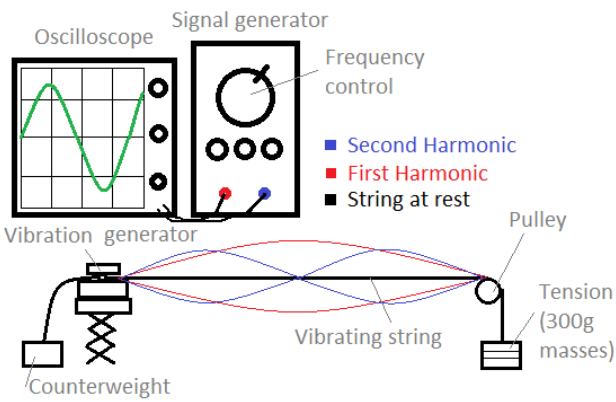
f: Frequency / Hz
v: Wave speed / ms^{-1}
λ: Wavelength / m
T: Tension / N
n: Number of harmonics
μ: Mass per unit length / gm^{-1}

$v = f\lambda$ (1)
$f = 1/T$ (2)
$v = \sqrt{(1000*T/\mu)}$; which is rearranged to $\mu/1000 = T/v^2$ (3)
$\lambda = 2/n$ (4)

Throughout this report, the above equations are referenced using the indices on the right.[2]

Method

Figure 1: Diagram of experimental setup



Setup:

The apparatus was set up as shown in *Figure 1*. The purpose of the pulley was to cause the wave to reflect back towards the vibration generator, which caused superposition and hence formed a standing wave on the string. The length of the string between the vibration generator and the pulley was constant and set at 1 metre. The free-hanging masses of 300g caused a tension of approximately 3 N in the string, which required the counterweight to balance and prevent the string from moving.

Practical:

1. The frequency of the signal generator was set to zero and slowly increased until the first harmonic could be observed on the string.
2. Using the oscilloscope, the time period of this wave was found, adjusting the settings if necessary.
3. The precise frequency was then calculated using *(2).*
4. The wavelength on the string was then calculated using *(4).*
5. The results from step 3 and 4 were combined using *(1)* to find the speed of the wave. This was recorded in a table.
6. The frequency on the signal generator was again steadily increased until the next harmonic was formed on the string. Steps 2 to 5 were repeated.
7. This was continued up until the 7th harmonic.

Collecting Results:

1. A graph of $1/\lambda$ against f was plotted. The uncertainty in f was considered.
2. Lines of best and worst fit were constructed to obtain the bounds on the value of the wave speed.
3. Finally, equation *(3)* was rearranged and used to obtain the value of μ.

Results

The results were analysed using Microsoft Excel[1]. The frequencies at which the harmonics occurred, the wavelengths and the calculated wave speed together with their uncertainties are shown in *Figure 2*:

Figure 2: Screenshot of Excel table of initial results

	Frequency / Hz: Attempt							
Harmonic #	#1	#2	#3	Mean F	λ / m	ΔF	εF	v / ms-1
1	9.1	9.62	9.4	9.37	2.00	0.26	2.77%	18.75
2	18.5	20	19.7	19.40	1.00	0.75	3.87%	19.40
3	27.8	27.8	27.8	27.80	0.67	0.10	0.36%	18.63
4	37	38.5	38.2	37.90	0.50	0.75	1.98%	18.95
5	38.9	47.6	47.8	47.70	0.40	4.45	9.33%	19.08
6	55.6	55.6	55.8	55.67	0.33	0.10	0.18%	18.37
7	63.3	65.8	64.9	64.67	0.29	1.25	1.93%	18.75

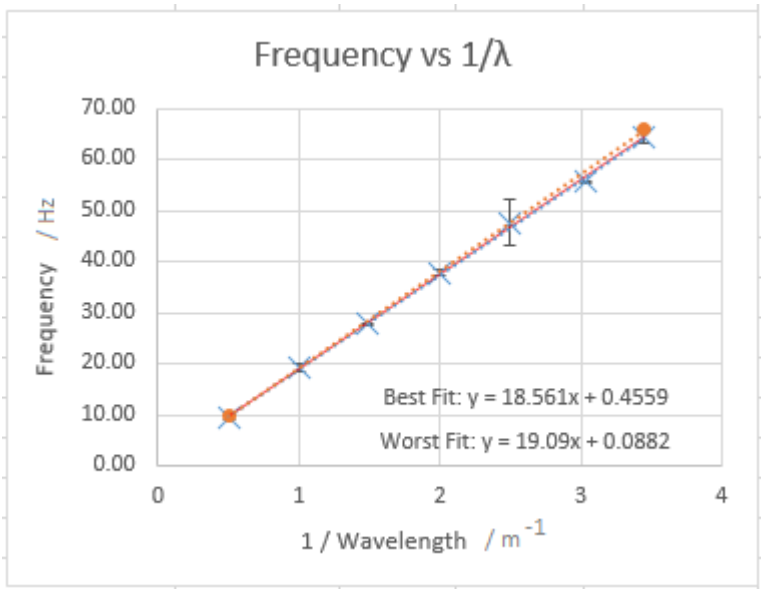
Anomalies, Errors and Uncertainties

Attempt #1 on the 5th harmonic was anomalous. This is likely due to either misreading the oscilloscope timebase or miscounting the number of harmonics on the wave as the result is very similar to that of the 4th harmonic. This result was not included in the subsequent calculations.

The absolute uncertainty is given by the half the range of the frequencies. For the 3rd harmonic, all attempts returned the same value, so the resolution of the oscilloscope was used instead.

A graph was plotted, presented in *Figure 3*:

Figure 3: Excel scatterplot of *1/λ* against *f:*



The gradient of this graph represents the wave speed, for which we obtained a value of $v = 18.6 \pm 0.5\ ms^{-1}$, a range of 2.7%.

On substitution of v into equation (3), we obtained a value of $\mu = 8.51 \pm 0.44\ gm^{-1}$, a range of 5.3%.

Conclusions

The obtained value $\mu = 8.51 \pm 0.44\ gm^{-1}$ is within the actual measured value of $8.68\ gm^{-1}$, and differs by approximately 2%. Hence, we consider this experiment a success as the correct value had been obtained to within these bounds.

References

[1] *Microsoft Excel*, B. Gates, 2019: https://www.office.com/launch/excel?auth=2

[2] *Some Guidelines on Report Writing,* R. Smith & D. Dagel, 2008